

Basis-Dependent Sparsity and Energy Recovery in a Strongly Correlated [2Fe–2S] Active Space

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Orbital rotations redistribute determinant amplitudes without changing a state, so coefficient weight need not certify energy. For the CAS(30e, 20o) [2Fe–2S] Hamiltonian, a Davidson Ritz vector gives $E = -116.60560912042631 E_h$, $184 \mu E_h$ below reported HCI, with residual $6.9016 \times 10^{-7} E_h$. Natural orbitals raise the top-512 norm from 0.191 to 0.769, but leave $169 m E_h$ error. Fixed-support and Hamiltonian-guided controls separate basis, coefficient, and support effects. Energy compressibility requires Hamiltonian contraction, not fidelity alone.

The compression of a correlated electronic state is inseparable from the choice of representation. Selected configuration-interaction (SCI) methods construct compact determinant spaces using estimates of Hamiltonian-mediated importance [1–3], whereas state-sampling workflows may first rank observed configurations by squared coefficient and then diagonalize the Hamiltonian in the retained subspace [4]. These operations answer different questions. The K largest coefficients give the best K -term approximation in Hilbert-space norm for a fixed orbital basis, but neither that support nor its inherited coefficients are guaranteed to minimize the energy. Moreover, an active–active orbital rotation leaves the physical state unchanged while generally redistributing its determinant amplitudes.

This distinction is especially important for iron–sulfur clusters, whose coupled high-spin centers generate dense multireference manifolds and severe entanglement [5, 6]. The $[\text{Fe}_2\text{S}_2(\text{SCH}_3)_4]^{2-}$ model in a CAS(30e, 20o) active space is consequently used to assess DMRG, neural wave functions, SCI, and quantum-classical methods [4, 6, 7]. Here we first obtain a lower explicit variational energy for this Hamiltonian. We then use the saved state to ask a representation-resolved question: how do orbital adaptation, coefficient relaxation, and Hamiltonian-guided support selection change the energy attainable with a finite determinant expansion?

Hamiltonian and Ritz calculation.— We consider 30 electrons in 20 orthonormal spatial orbitals, restricted to $M_S = 0$, with

$$\hat{H} = \sum_{pq\sigma} h_{pq} a_{p\sigma}^\dagger a_{q\sigma} + \frac{1}{2} \sum_{pqrs\sigma\tau} g_{pqrs} a_{p\sigma}^\dagger a_{r\tau}^\dagger a_{s\tau} a_{q\sigma}. \quad (1)$$

The spin-string coefficient tensor has shape $\binom{20}{15} \times \binom{20}{15} = 15,504^2$, or 240,374,016 real amplitudes. We use the archived benchmark FCIDUMP [8]. A second FCIDUMP distributed with the QMB models [6, 9] is related to it by an orthogonal orbital rotation: the transformed one-electron tensor agrees to $3.55 \times 10^{-14} E_h$ elementwise, and the complete two-electron tensor has root-mean-square and maximum discrepancies $3.08 \times 10^{-15} E_h$ and $1.21 \times 10^{-13} E_h$, respectively.

We minimize the Rayleigh quotient in the complete configuration-interaction space *within* this active space and spin sector,

$$E[\mathbf{c}] = \frac{\mathbf{c}^\top \mathbf{H} \mathbf{c}}{\mathbf{c}^\top \mathbf{c}}, \quad \rho(\mathbf{c}) = \frac{\|\mathbf{H} \mathbf{c} - E[\mathbf{c}] \mathbf{c}\|_2}{\|\mathbf{c}\|_2}, \quad (2)$$

using a diagonally preconditioned, restarted Davidson iteration [10] and matrix-free Slater–Condon contractions in PySCF [11, 12]. Atomic checkpoints permit wider-subspace restarts without replacing the explicit state by extrapolated energies. The final production stage used a 64-vector maximum subspace, 160 updates, an energy threshold $10^{-13} E_h$, and residual target $10^{-8} E_h$. The full staged simulation, initialization, environment, and executable commands are given in the Supplemental Material [13].

The saved vector has squared norm 0.999999999999999 and direct Rayleigh energy and residual

$$\begin{aligned} E_\star &= -116.60560912042631 E_h, \\ \rho_\star &= 6.9016 \times 10^{-7} E_h. \end{aligned} \quad (3)$$

This energy is approximately $184 \mu E_h$ below the reported HCI value $-116.605425 E_h$ [14]; the precision shown for that entry does not support finer comparison. It lies in the rounding interval of the published DMRG value $-116.6056091 E_h$ [6, 9], so we claim agreement with, not improvement over, DMRG [Fig. 1(a)]. A fresh-process Hamiltonian application reproduces Eq. (3); a separate one- and two-particle density-matrix contraction agrees within $5.97 \times 10^{-13} E_h$, and $\langle \hat{S}^2 \rangle = 2.86 \times 10^{-9}$. The Davidson convergence flag is false because $\rho_\star$ exceeds the preset target [Fig. 1(c)]. Nevertheless, the explicit Rayleigh quotient is a variational upper bound to the lowest eigenvalue in the stated sector. We do not infer a two-sided error bar from the residual.

Representation-resolved compression.— We compare the benchmark orbital basis, denoted $b = \text{B}$, with an a posteriori natural-orbital basis $b = \text{N}$. Diagonalizing the spin-summed one-particle density matrix gives $\gamma U = U \text{diag}(n_p)$. We transform both the integrals and the complete CI tensor by the same U ; direct contractions before and after the rotation preserve the norm, energy,

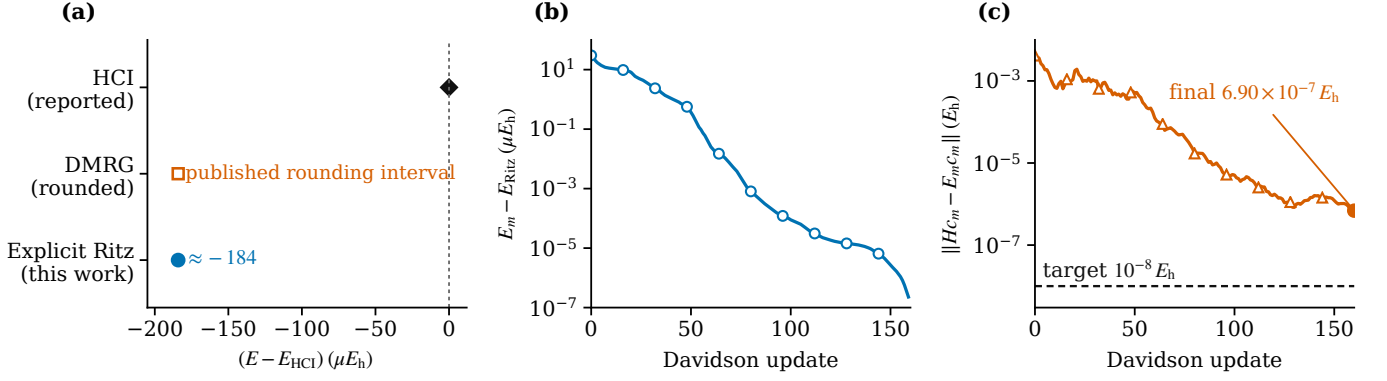


FIG. 1. **Explicit variational energy and numerical convergence.** (a) Energies relative to the reported HCI entry. The DMRG segment denotes only the interval that rounds to the published digits. (b) Positive Ritz-energy difference from the final saved state during the production stage. (c) Residual norm of the same iterates. The final point remains above the preset $10^{-8} E_h$ target.

and residual to 1.4×10^{-14} , $3.13 \times 10^{-13} E_h$, and $1.12 \times 10^{-16} E_h$, respectively. Thus the two coefficient tensors are representations of the same physical Ritz state, not separate electronic solutions.

For each representation, $P_K^{(b)}$ retains the K largest values of $|c_I^{(b)}|$. We define

$$\begin{aligned} W_K^{(b)} &= \langle c | P_K^{(b)} | c \rangle, \\ |\phi_K^{(b)}\rangle &= \frac{P_K^{(b)} | c \rangle}{\sqrt{W_K^{(b)}}}, \\ \Delta E_K^{(b)} &= \langle \phi_K^{(b)} | H | \phi_K^{(b)} \rangle - E_\star. \end{aligned} \quad (4)$$

Every projected energy and residual in Fig. 2 is obtained from a new full Hamiltonian contraction. No energy is estimated from discarded weight. The natural orbitals strongly concentrate the norm: at $K = 64$, W_K rises from 0.0755 to 0.4181; at $K = 512$, it rises from 0.1913 to 0.7685. The energetic improvement is real but not commensurate with the fidelity: the natural-orbital $K = 512$ projection remains 168.98 m E_h above $E_\star$, with residual $0.473 E_h$. At $K = 4,194,304$, the natural-orbital projection increases W_K from 0.985678 to 0.999776 and reduces ΔE_K from 17.705 to 0.3488 m E_h ; its residual nevertheless remains $2.72 \times 10^{-2} E_h$.

To isolate what amplitude pruning leaves unresolved, we perform two $K = 512$ controls in the benchmark basis. The inherited top-512 coefficients give $-116.0003434451 E_h$. Exact diagonalization on that *same* support lowers the energy by 145.97 m E_h , to $-116.1463085705 E_h$. An independently constructed, Hamiltonian-guided 512-determinant support, followed by exact subspace diagonalization, gives $-116.3704626758 E_h$. Its advantage over the inherited top-512 state is 370.12 m E_h , but it remains 235.15 m E_h above $E_\star$. A fermionic-operator reference implementation and a PySCF full-space embedding agree on both rediagonalized

energies within $3.2 \times 10^{-13} E_h$. Their projected-support residuals are below $2.5 \times 10^{-13} E_h$, whereas their full-space residuals are 0.558 and 0.215 E_h . These controls establish coefficient stationarity only on their stated supports; they neither identify a globally optimal support nor bound a fully adaptive SCI calculation.

The a posteriori natural-orbital top-512 projection is lower still, $-116.4366261444 E_h$, by 66.16 m E_h relative to the Hamiltonian-guided benchmark-basis control. Rediagonalizing its unchanged support gains another 49.634 m E_h , giving $-116.486260278799 E_h$. The projected residual is $1.68 \times 10^{-13} E_h$, but the full-space residual is still $0.311254 E_h$: coefficient stationarity does not remove coupling to excluded determinants. These comparisons show how strongly the one-particle representation can affect a fixed determinant budget. They are not a constructive sparse-solver benchmark, because the natural orbitals were obtained from the complete Ritz state.

The missing energetic contribution can be resolved without reference to the arbitrary zero of energy. Let $A = H - E_\star I$, $p = P_K c$, $q = (I - P_K)c$, and $W = \langle p | p \rangle$. Since $\langle c | A | c \rangle = 0$,

$$\Delta E_K = -\underbrace{\frac{2 \operatorname{Re} \langle p | A | q \rangle}{W}}_{C_K} + \underbrace{\frac{\langle q | A | q \rangle}{W}}_{T_K}. \quad (5)$$

Using the directly contracted $E_\star$, E_K , and $\langle c | H | \phi_K \rangle$, we algebraically reconstruct both terms; their maximum floating-point recombination error is $6.1 \times 10^{-14} E_h$. In the benchmark representation at $K = 512$, the removed retained-tail coupling contributes $C_K = 1.2105313 E_h$, while the tail quadratic term contributes $T_K = -0.6052656 E_h$; their cancellation leaves the $0.60527 E_h$ error of the normalized projection. The same pattern persists across the evaluated supports [Fig. 2(d)]. Large discarded contributions can therefore cancel in the full state even when the discarded norm is small; W_K alone contains neither matrix elements nor their signs.

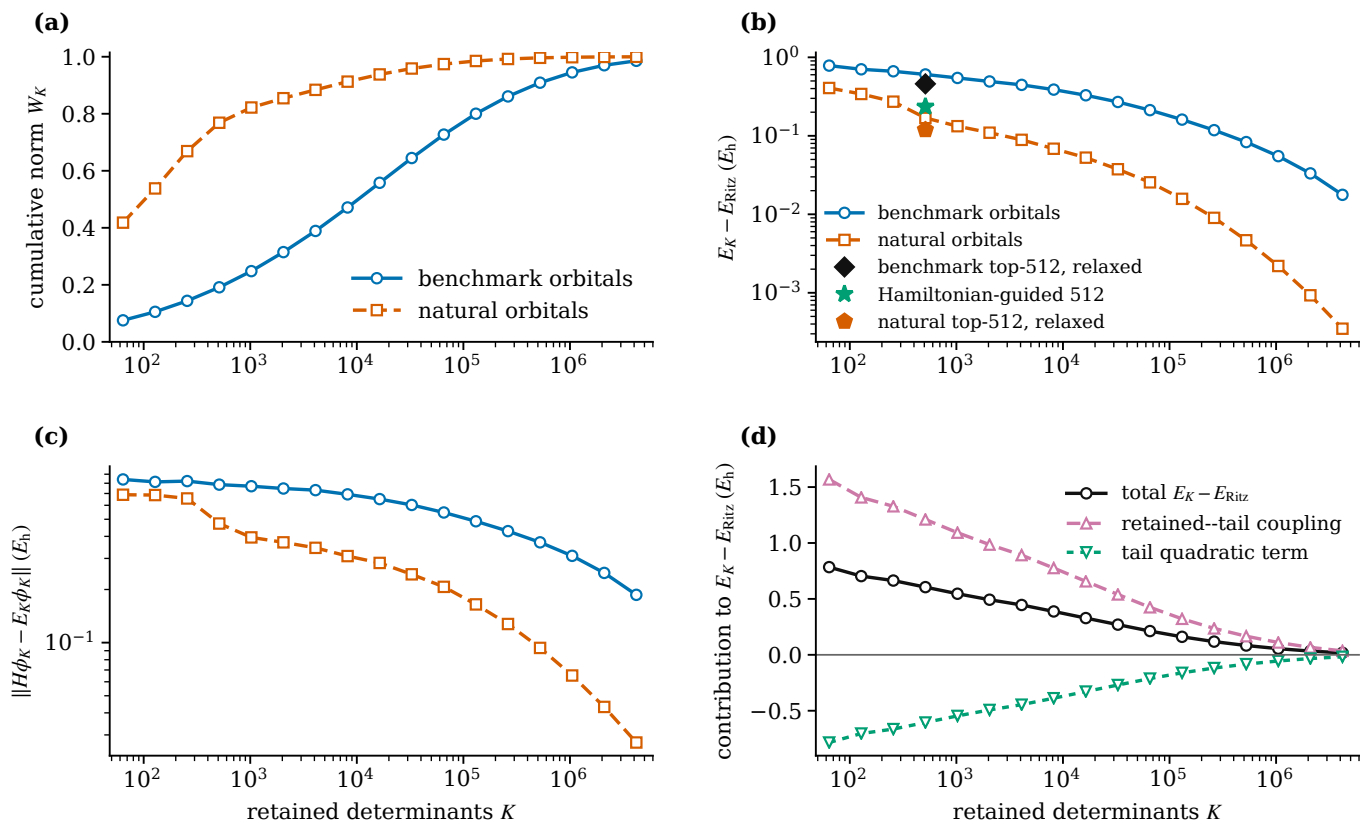


FIG. 2. **Orbital adaptation, coefficient relaxation, and support choice.** (a) Cumulative norm of amplitude-ranked projections in the benchmark and natural-orbital representations. (b) Directly contracted energy errors. The diamond and pentagon are exact ground states on the benchmark- and natural-basis top-512 supports, respectively; the star is an independently constructed Hamiltonian-guided 512-determinant support in the benchmark basis. (c) Full-space residuals of the fixed-coefficient projections. (d) Gauge-invariant decomposition of the benchmark-basis error into removed retained-tail coupling and tail quadratic terms. The latter is negative; their sum is the positive projected-state error.

Our conclusion is deliberately narrower than a claim about all SCI methods. Natural orbitals substantially improve determinant sparsity for this state, and Hamiltonian-based coefficient and support optimization substantially lower sparse-state energies. Yet in both tested representations the mapping from retained norm to energy error must be established by Hamiltonian contraction. For sampling-based subspace methods, this diagnoses the amplitude-ranking step, not the subsequent subspace diagonalization. More broadly, fidelity, support choice, coefficient stationarity, and full-space stationarity are distinct properties and should be reported separately.

In summary, a reproducible Davidson calculation yields an explicit CAS(30e, 20o) variational energy below the reported HCI entry and consistent with rounded DMRG. Exact orbital transformation and sparse controls then show that orbital adaptation changes how compact the coefficient vector appears without turning amplitude weight into an energetic certificate. The result supplies both a lower classical reference and a representation-aware protocol for testing sparse classical and quantum ansatzes.

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